3. (a) (i) Friction f between the tyres and the road. 1A

$$f = \frac{mv^2}{r}$$ 1M

$$8000 = \frac{1200\,v^2}{45\text{ m}}$$

$$v = 17.3\text{ m s}^{-1}$$ 1A 3

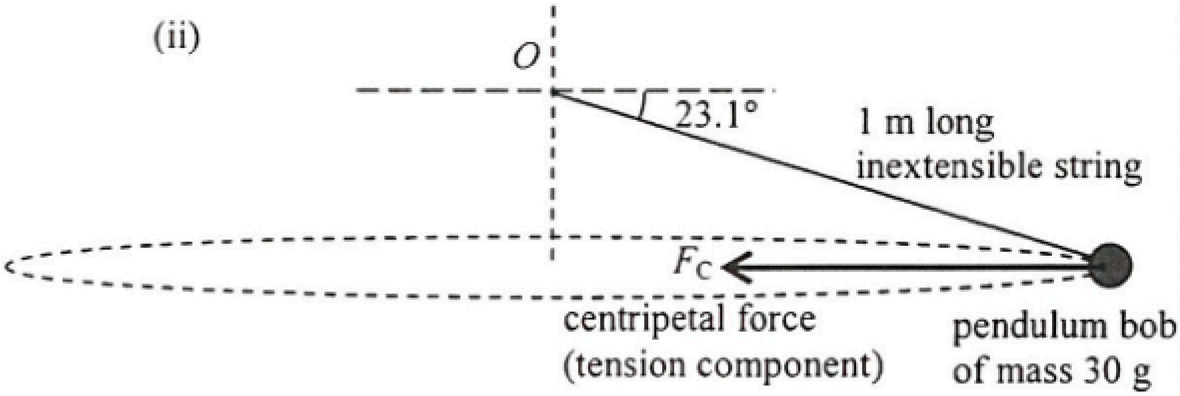
(ii) Smaller 1A

For the same f, $v^2 \propto r$; when r decreases v decreases. 1A 2

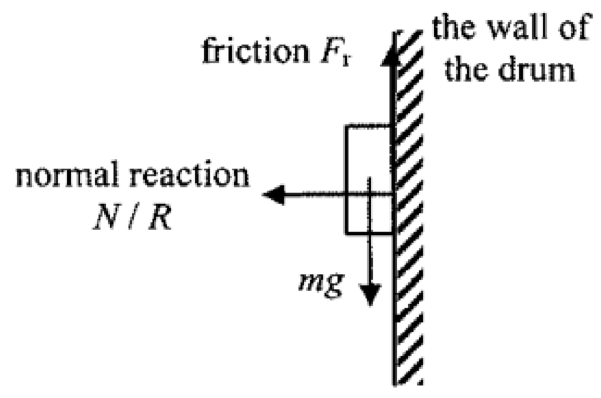
(b) (Max) friction / coefficient of friction reduced, 1A

not enough to provide the centripetal force / acceleration required for circular motion.

Or Tracking or allowed speed lowered. 1A 2

4. (a) (i)	Rotation rate $= \dfrac{\omega}{2\pi} = \dfrac{5.0}{2\pi}$ $= 0.795775\ (\text{rev s}^{-1}) \approx 0.80\ (\text{rev s}^{-1})$	1M/1A	Accept: $(0.79 \sim 0.80)$ rev s^{-1}
		1	
(ii)			
	F_C correctly indicated.	1A	
	$F_C = mr\omega^2$ $= (0.03)(1 \times \cos 23.1°)(5.0)^2$ $= 0.689866\text{ N} \approx 0.690\text{ N}$ ($F_C = 0.7033402\text{ N} \approx 0.703\text{ N}$ for $g = 10\text{ m s}^{-2}$)	 1M 1A	
	OR $T\cos\theta = F_C$ and $T\sin\theta = mg$ $F_C = \dfrac{mg}{\sin\theta}\cos\theta = 0.689866\text{ N} \approx 0.690\text{ N}$	 1M 1A	
		3	
(iii)	Horizontal component of tension provides the centripetal force, thus tension is larger than the centripetal force.	1M 1A	
	OR $T\cos\theta = F_C$ $\Rightarrow T > F_C$	1M 1A	
	OR $T\sin\theta = mg$ $\Rightarrow T\ (0.750\text{ N}) > F_C\ (0.690\text{ N})$	1M 1A	
		2	
(b) (i)	The gravitational force is always perpendicular to the moon's motion / displacement / velocity, thus no work is done on the moon by this force (k.e. unchanged)	1A 1A	
		2	
(ii)	(The claim is incorrect) as, by Newton's third law of motion, gravitational force of the same magnitude (but in opposite direction) is acting on the Earth by the moon.	1A 1A	Accept: action and reaction pair
		2	

4. (a) (i)



2A

2

(ii) Normal reaction (force acting on drum/wall of the drum by the object) 1A

1

(b) (i) $\omega = 78.5$ rad s^{-1}

rotation speed $= \frac{78.5}{2\pi} \times 60$ 1M

$= 749.619782 \approx 750$ (rev min^{-1}) 1A

Or

$T = \frac{2\pi}{\omega} = 0.0800441$ s and

$f = \frac{1}{T} = 12.493663$ rev s^{-1} 1M

$= 749.619782 \approx 750$ (rev min^{-1}) 1A

2

(ii) centripetal force

$= m\omega^2 r$

$= (2 \times 10^{-5})(78.5)^2(0.25)$ 1M

$= 0.03081125$ N ≈ 0.0308 N 1A

2

(iii) $v = r\omega$

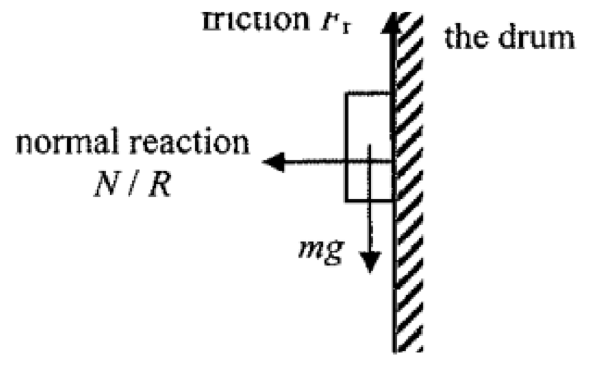
$= 0.25 \times 78.5$ 1M

$= 19.625 \approx 19.6$ m s^{-1} 1A

2

(iv) I 1A

1



2A

2

(ii) Normal reaction (force acting on drum/wall of the drum by the object) 1A

1

(b) (i) $\omega = 78.5$ rad s^{-1}

rotation speed $= \frac{78.5}{2\pi} \times 60$ 1M

$= 749.619782 \approx 750$ (rev min^{-1}) 1A

Or

$T = \frac{2\pi}{\omega} = 0.0800441$ s and

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(ii) centripetal force

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$= 0.03081125$ N ≈ 0.0308 N 1A

2

(iii) $v = r\omega$

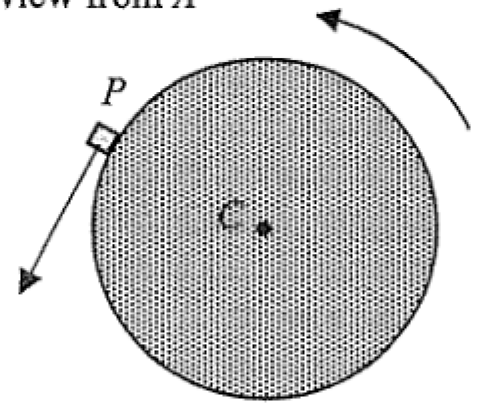
$= 0.25 \times 78.5$ 1M

$= 19.625 \approx 19.6$ m s^{-1} 1A

2

(iv) I 1A

1

4.	(a)	(i)	$F = m\omega^2 r = 0.020 \times 6^2 \times 0.50$	1M
			$= 0.36\ \text{N}$	1A
				2
		(ii)	Top view from A 	1A
				1
	(b)	(i)	same angular speed	1A
				1
		(ii)	different / smaller magnitude of centripetal acceleration ($a_Q < a_P$)	1A
			as Q takes a different / smaller radius ($r_Q < r_P$)	1A
				2